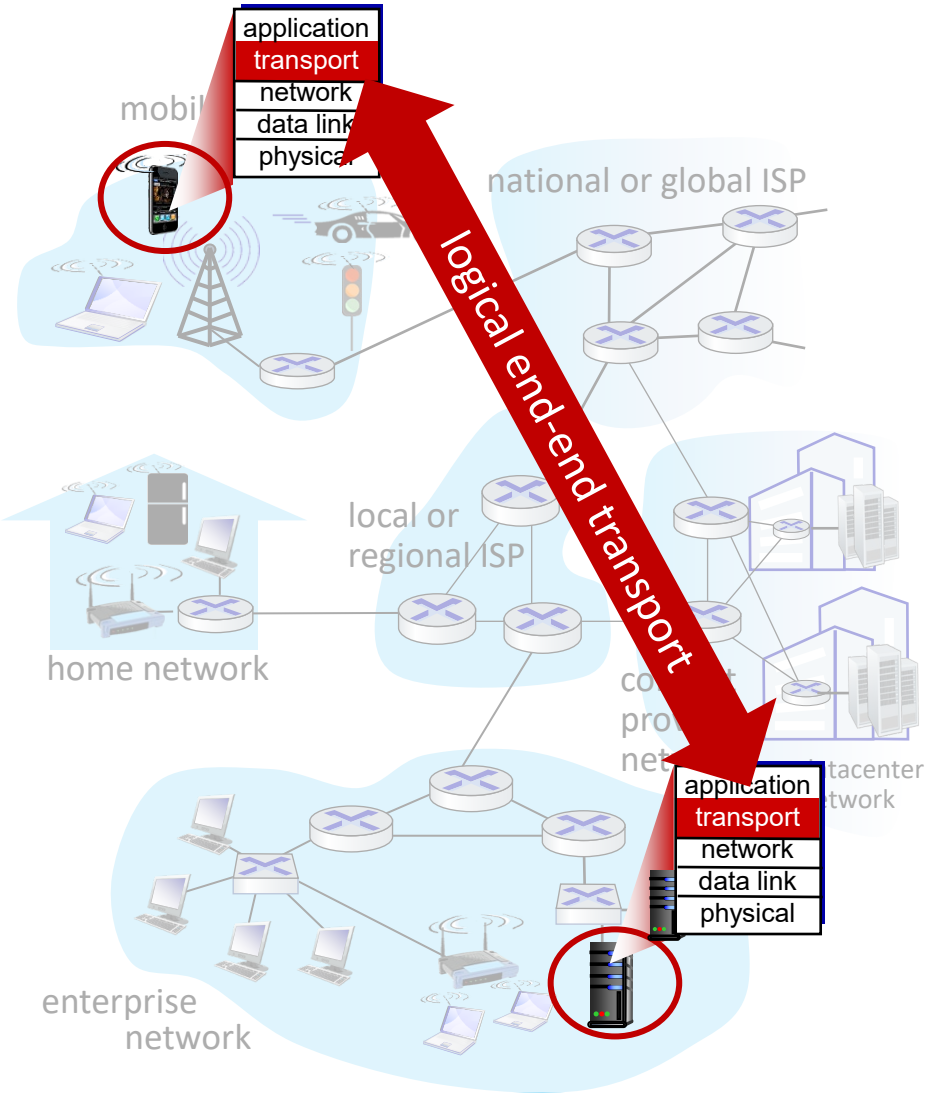
# Lecture 10-Part 2

- Transport Layer Services
- Multiplexing and Demultiplexing

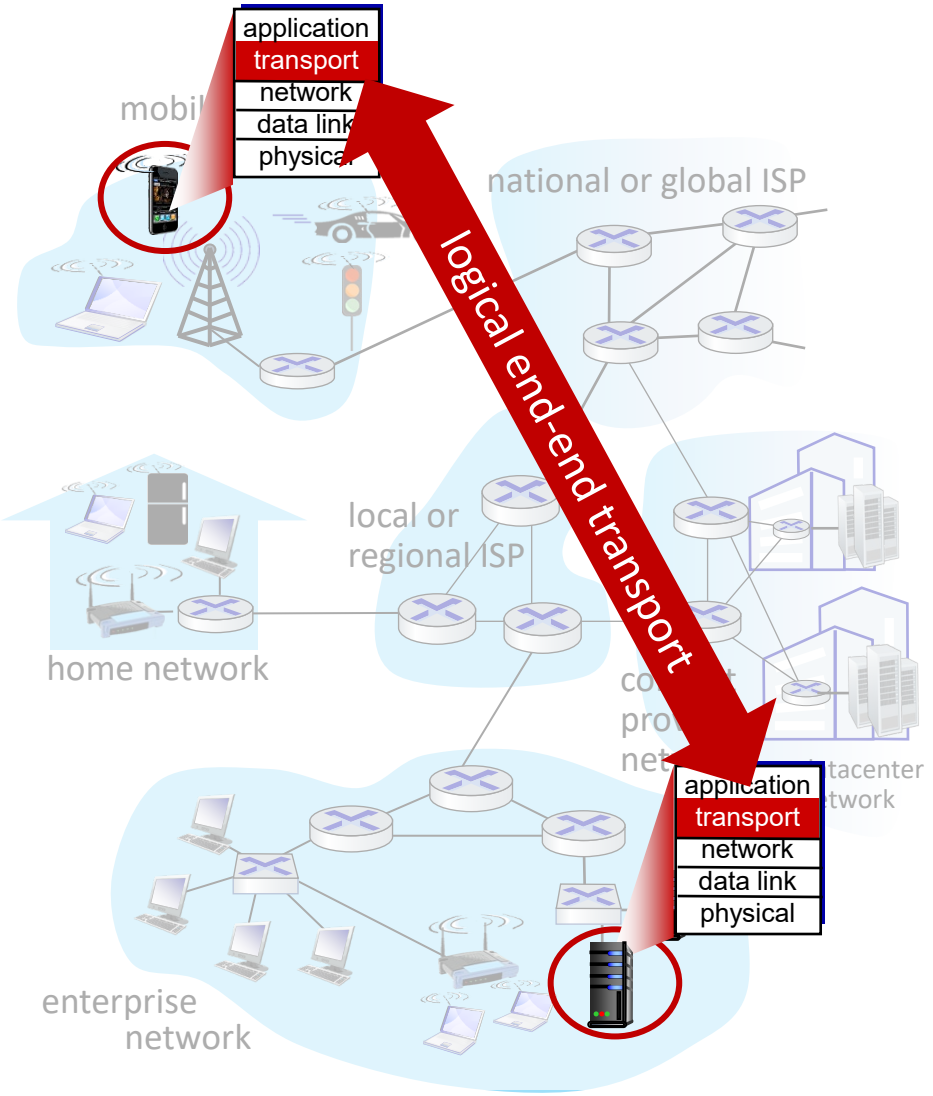
# Transport services and protocols

- Provides *logical communication* between ***application processes*** running on different hosts
- Transport protocols actions in end systems:
  - ***sender***: breaks app messages into *segments*, passes to network layer
  - ***receiver***: reassembles segments into messages, passes to application layer
- Two **popular** transport protocols available to Internet applications
  - TCP, UDP



# Two principal Internet transport protocols

- **TCP:** Transmission Control Protocol
  - reliable, in-order delivery
  - congestion control
  - flow control
  - connection setup
  - RFC 793 (original doc: link)
- **UDP:** User Datagram Protocol
  - unreliable, unordered delivery
  - no-frills extension of “best-effort” IP
  - RFC 768 (originall doc: link)
- Services not available:
  - delay guarantees
  - Bandwidth (data rate) guarantees

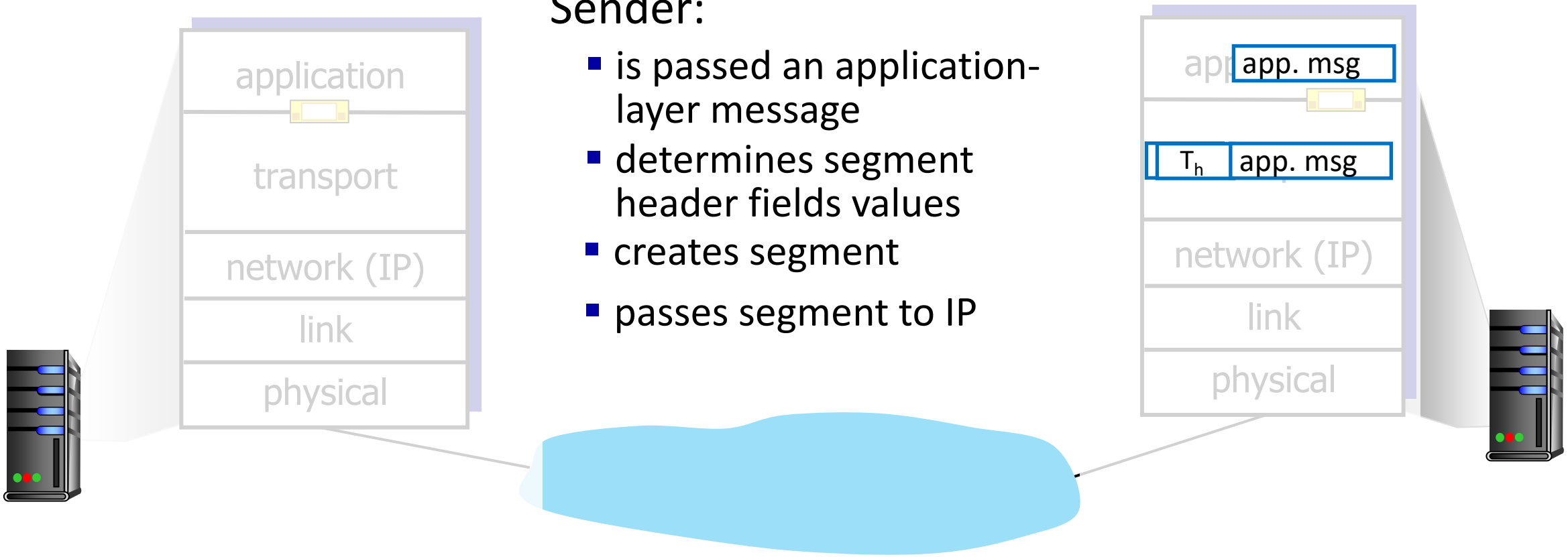


# Transport vs. Network layer

- Transport layer*:* logical communication between *processes*
  - relies on and enhances network layer services

- Network layer: logical communication between *hosts*
- *It May*
  - Drop messages (datagrams)
  - Re-order messages
  - Delivers duplicate copies of a given message
  - Limits messages to some finite size
  - Delivers messages after long delays

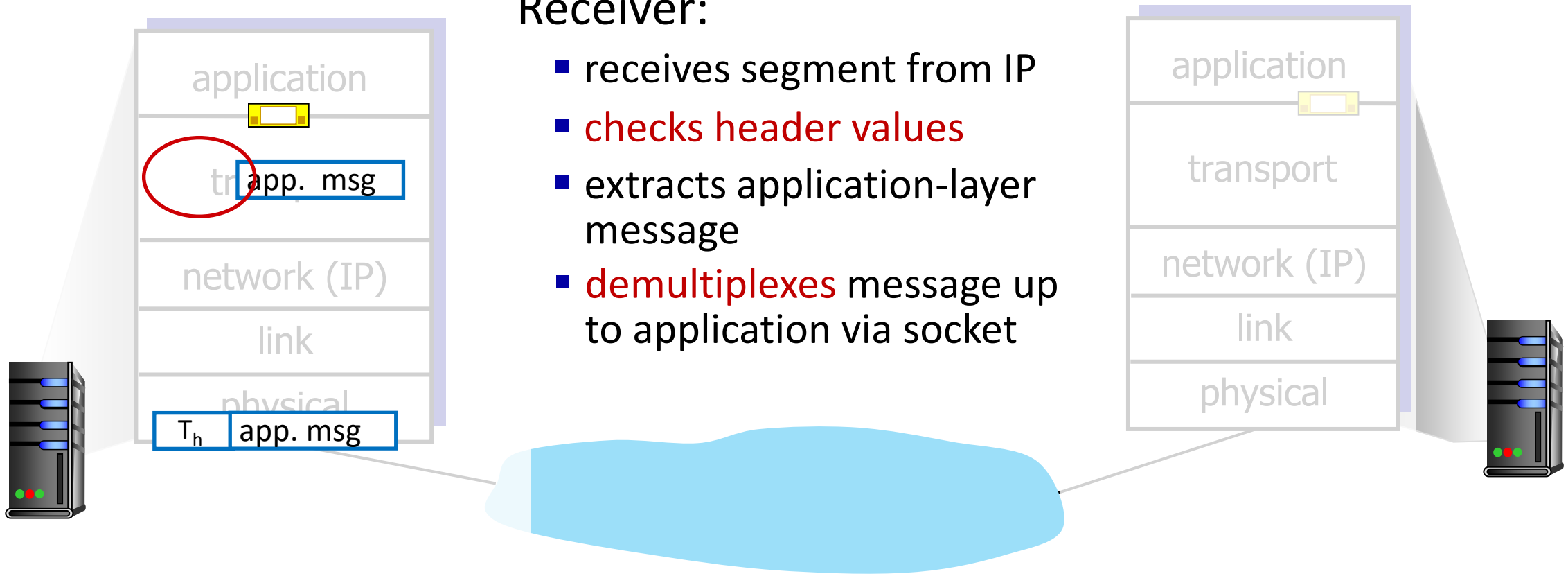
# Transport Layer Actions




Sender:

- is passed an application-layer message
- determines segment header fields values
- creates segment
- passes segment to IP

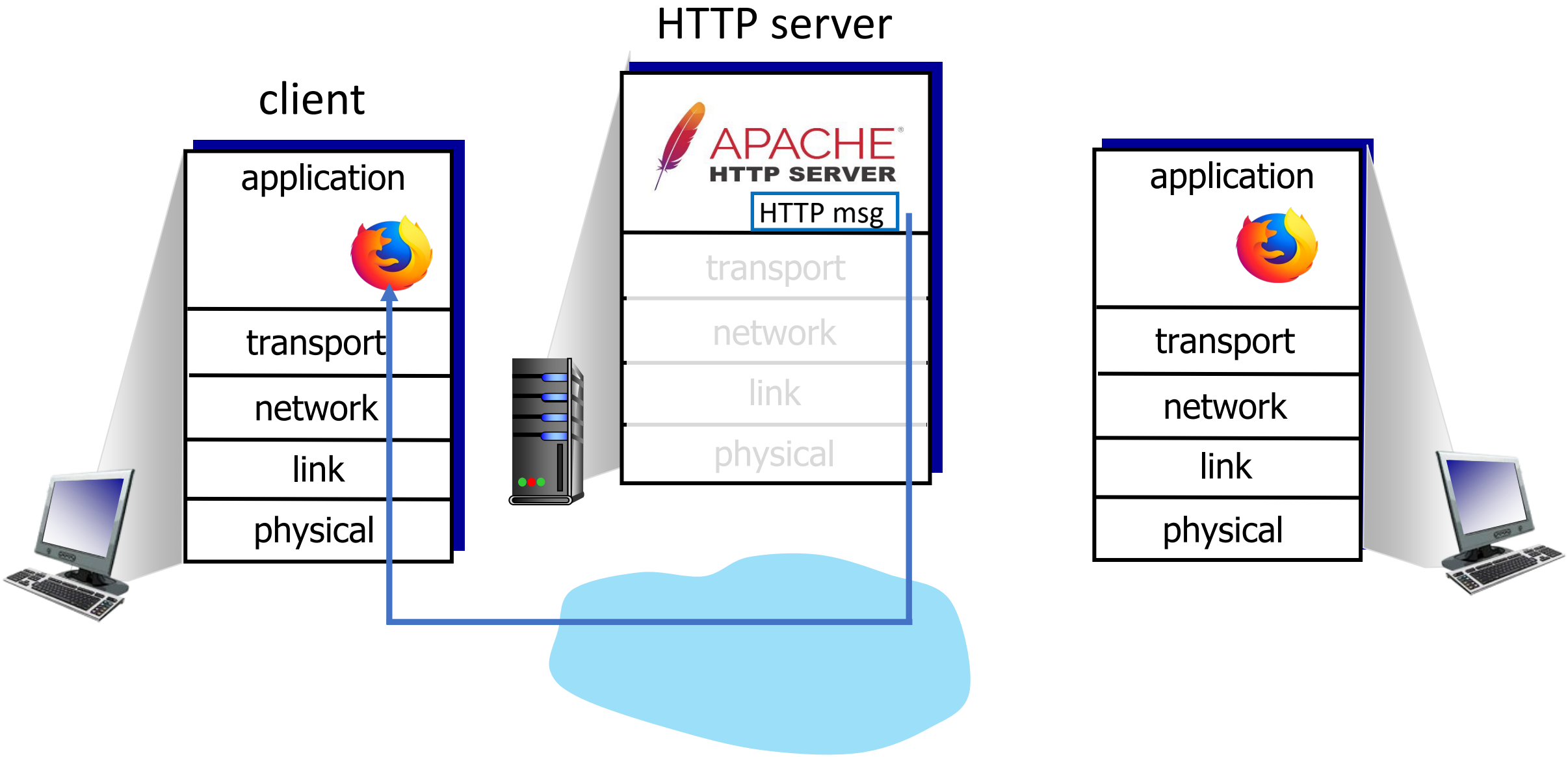
# Transport Layer Actions



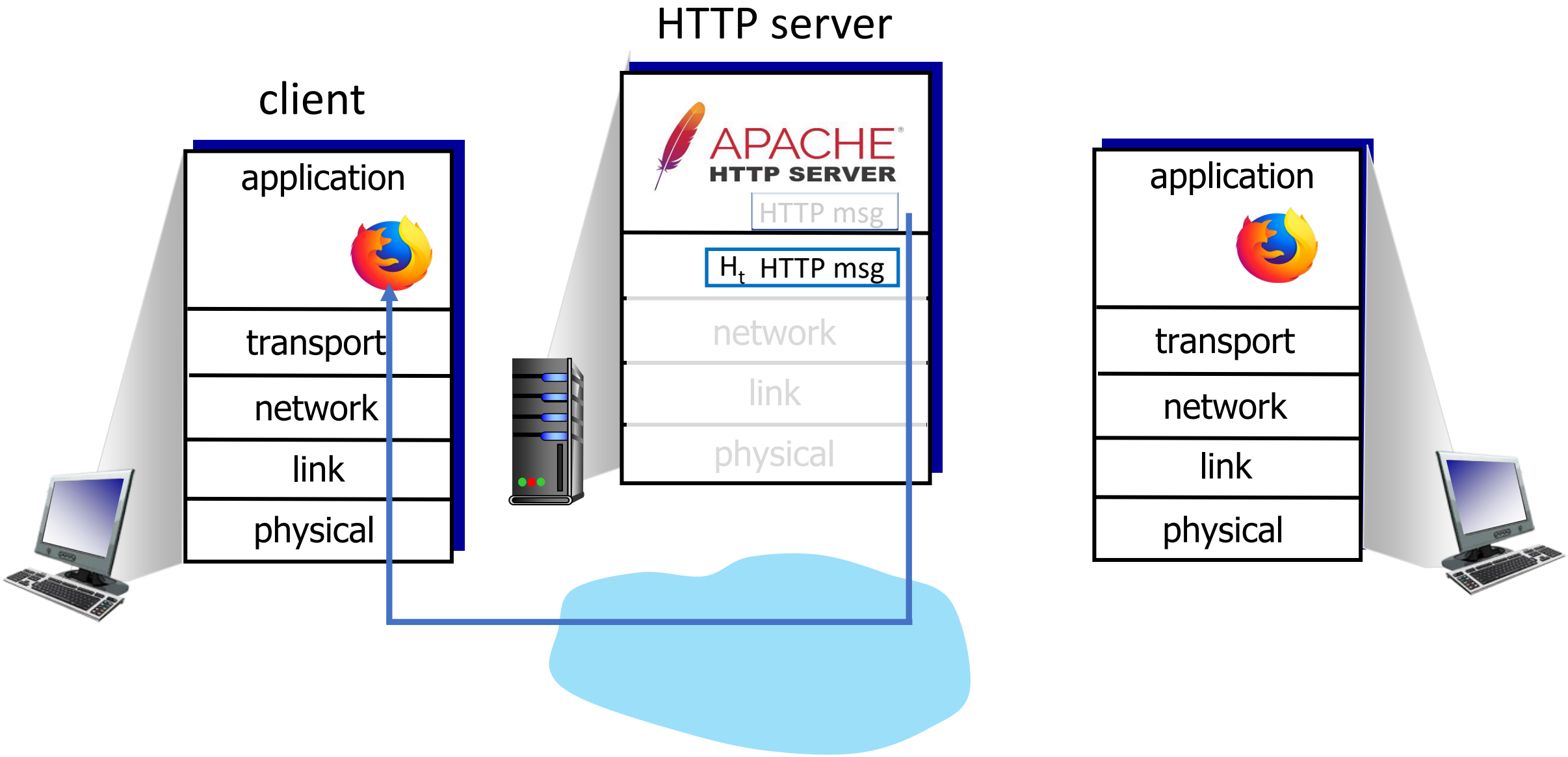

Receiver:

- receives segment from IP
- checks header values
- extracts application-layer message
- demultiplexes message up to application via socket

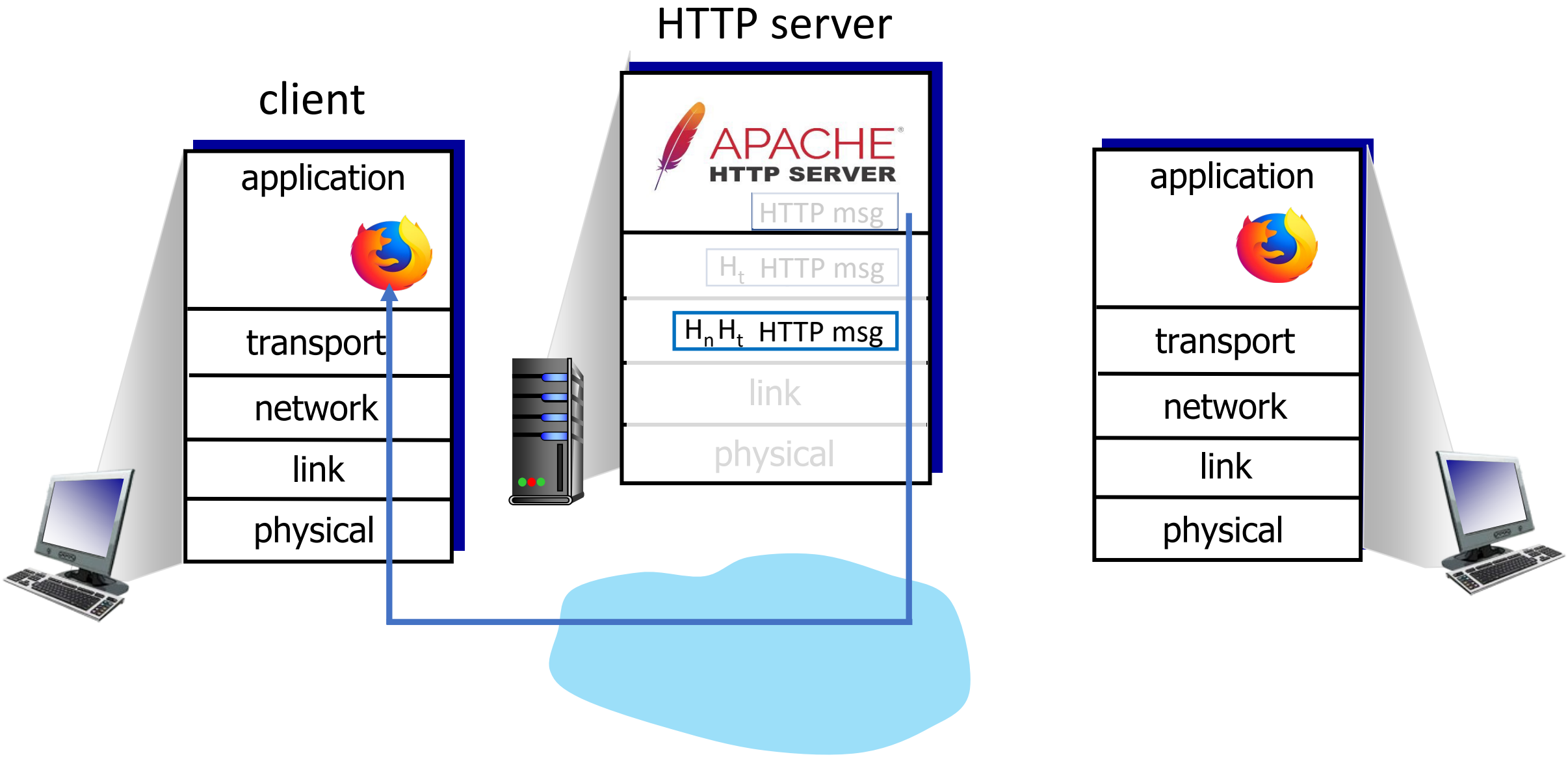
# Multiplexing and demultiplexing: Problem Statement



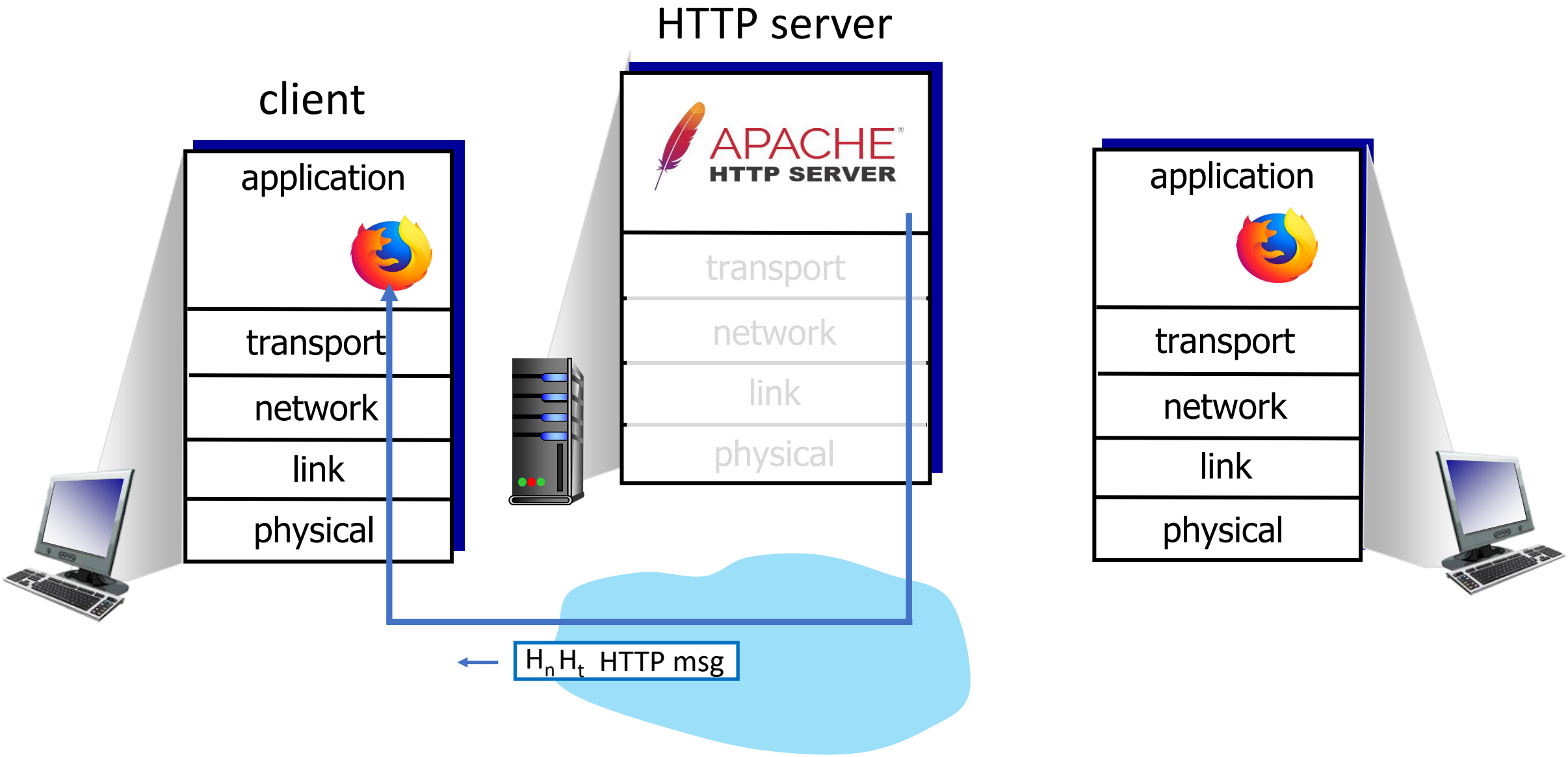
# Problem Statement



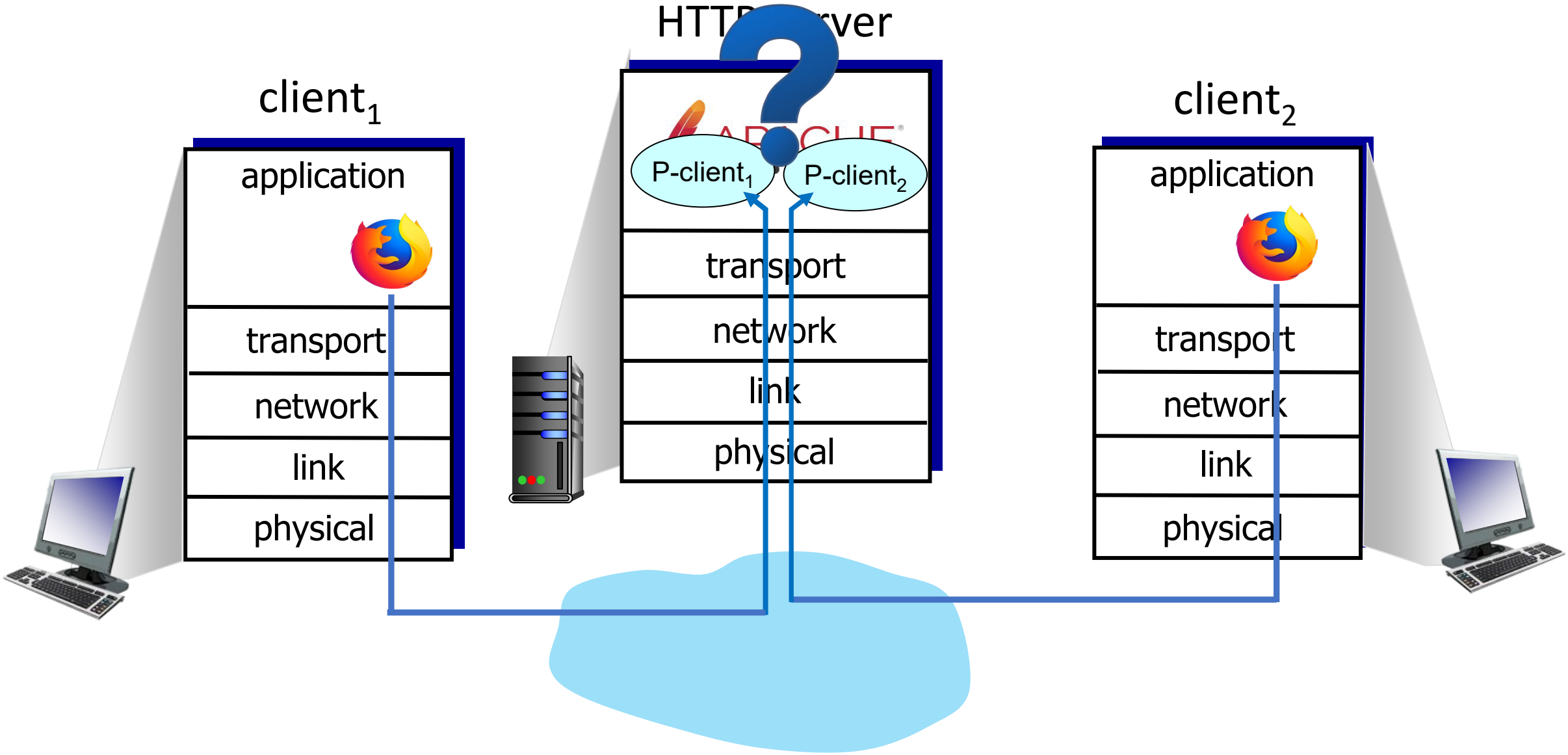
# Problem Statement



# Problem Statement

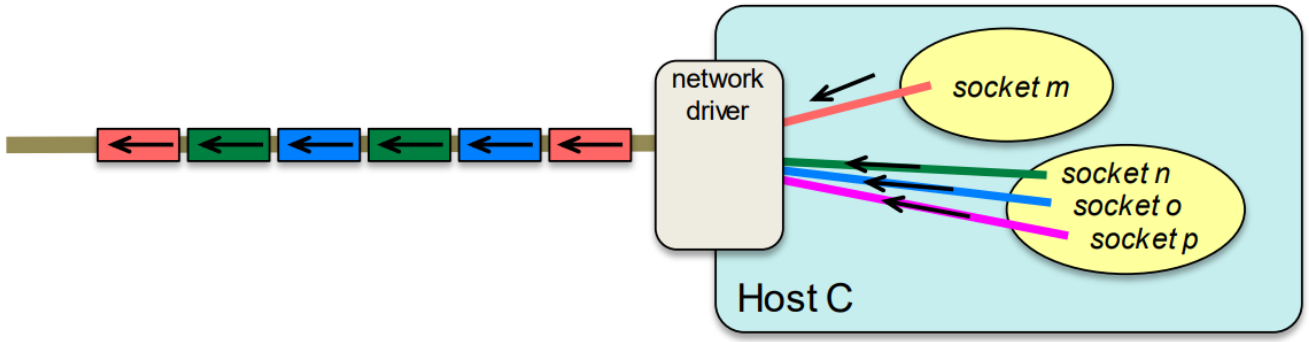


# Problem Statement

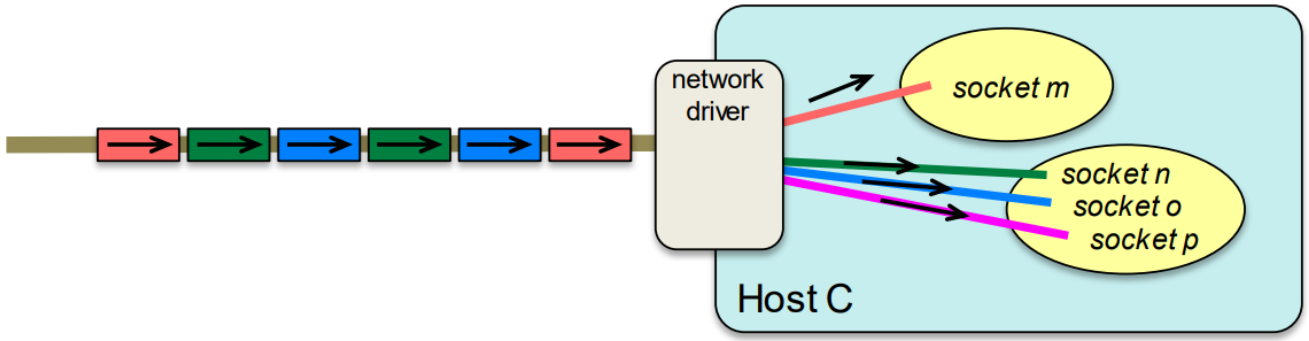


# Multiplexing & Demultiplexing Concepts

## Multiplexing




## Demultiplexing



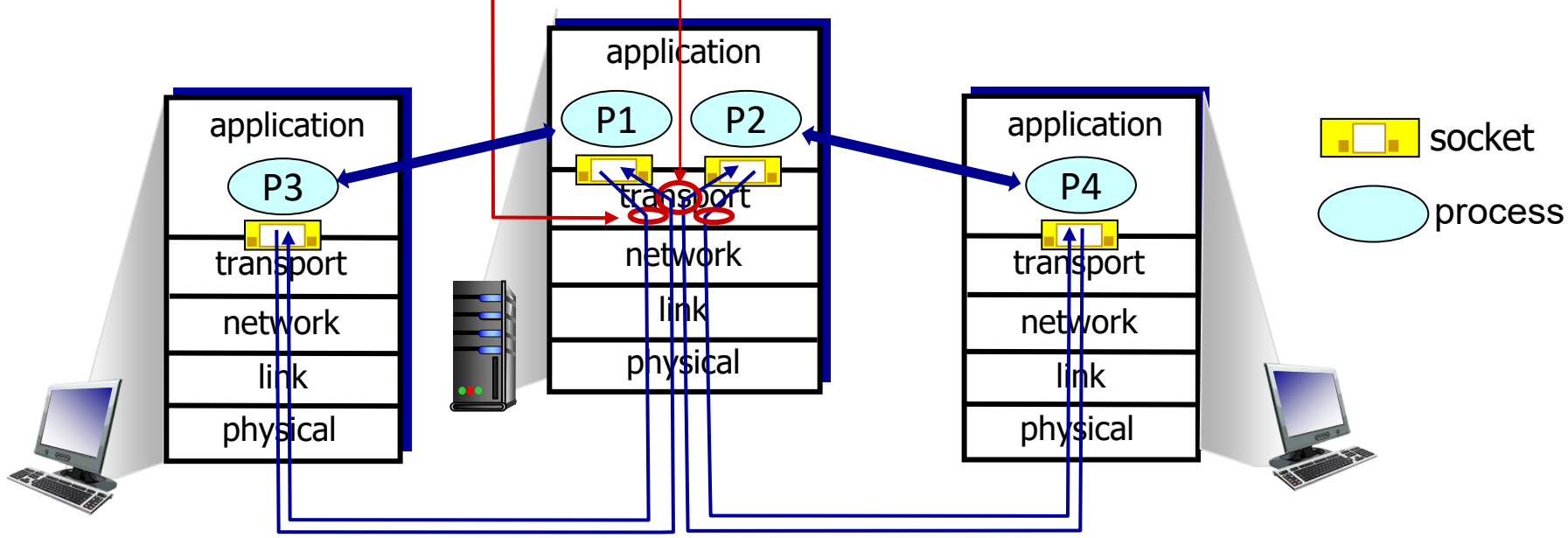
# Multiplexing/demultiplexing

*multiplexing at sender:*

handle data from multiple sockets, add transport header (later used for demultiplexing)

*demultiplexing at receiver:*

use header info to deliver received segments to correct socket



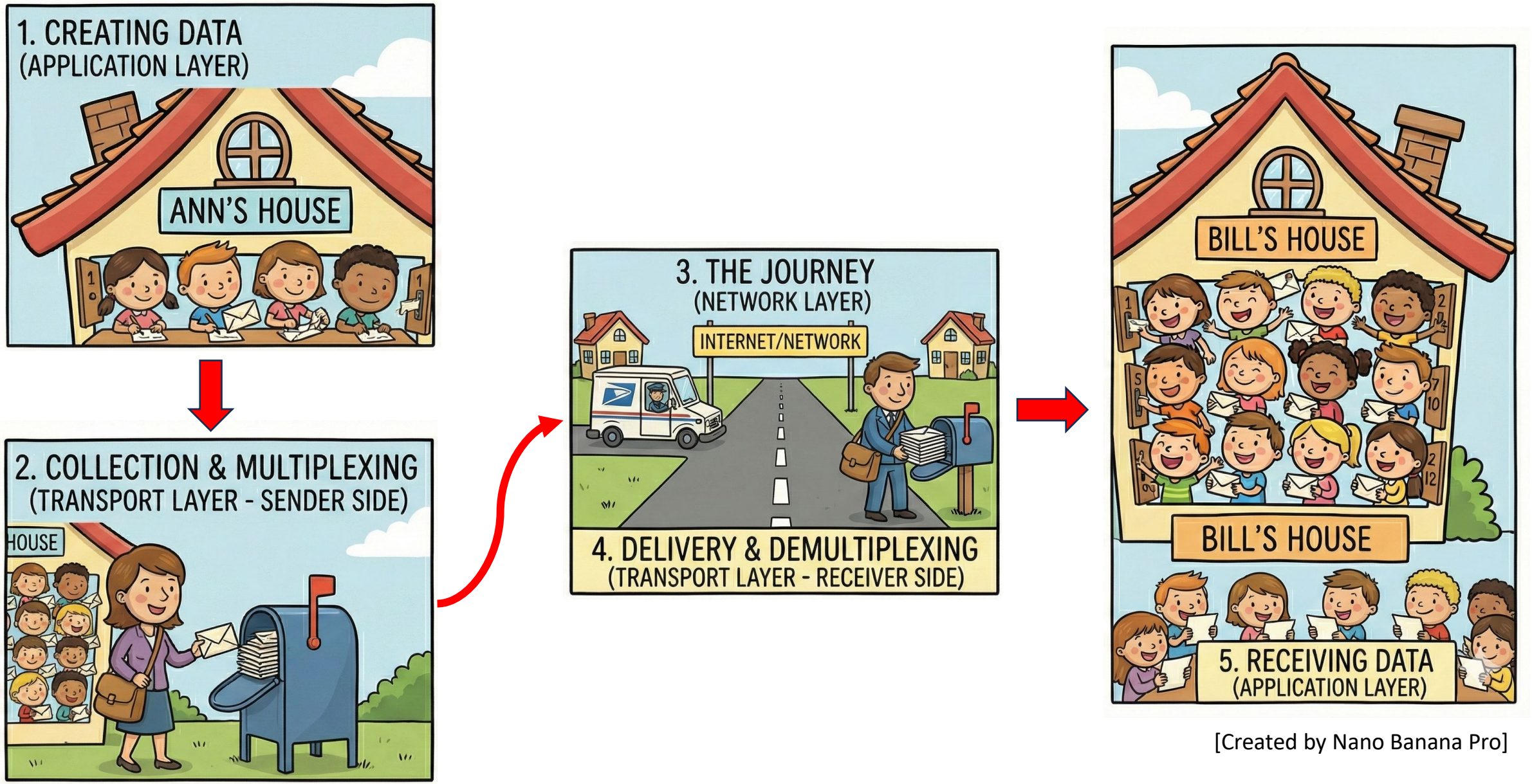
# *Household analogy*

***Scenario**: 12 kids in Ann's house sending letters to 12 kids in Bill's house;*

- Ann collects letters from each kid's room and places them in mailbox.
- Bill picks up letters from mailbox and delivers to each kid's room.

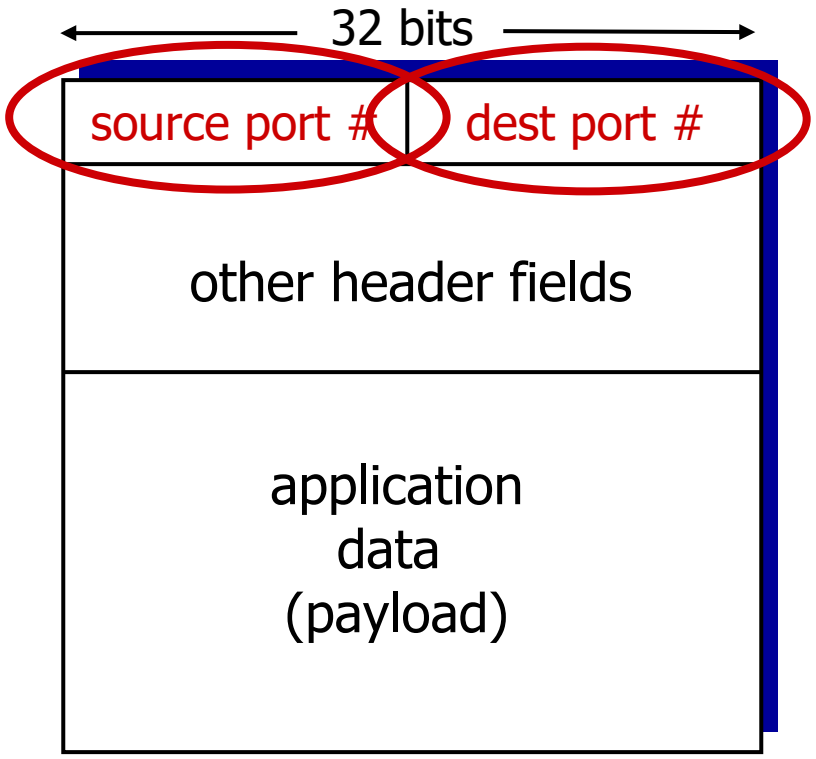
| Household Component | Corresponding Networking Component |
|---|---|
| **Houses** | **Hosts** |
| **Kids** | **Processes** |
| **Letters in envelopes** | **App messages** |
| **Ann and Bill** (multiplex and demultiplex to in-house siblings) | **Transport Protocol** |
| **Postal service** | **Network-layer protocol** |

# Fun Demonstration




[Created by Nano Banana Pro]

# TCP/UDP Segment Format




TCP/UDP segment format